Compression and Expansion

How Physicists and Biologists Describe the Same Phenomena

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Abstract

Physics and biology both aim to explain the natural world, yet the character of their explanations differs fundamentally. Physics compresses phenomena into invariant relations: minimal equations that simultaneously describe and explain, admitting mathematisation precisely because the systems under study are low-dimensional and closed under conservation laws. Biology expands the same phenomena into mechanistic narratives: causal chains distributed over time, structure, and context. This paper examines six phenomena through both lenses, arguing that the distinction is not one of sophistication but of *closure*. Where a system's degrees of freedom can be isolated and its symmetries identified, description collapses into algebra. Where they cannot, explanation must be unfolded into process.

Introduction: Two Modes of Explanation

Consider a physicist asked to explain why a magnet slows as it falls through a copper tube. The answer is brief: Faraday's law, Lenz's law, conservation of energy. A single line of mathematics suffices:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

The minus sign does all the explanatory work. It encodes opposition, causation, and energy conservation simultaneously.

Now consider how a biologist, trained in the expansion of mechanism, would approach the same phenomenon. They would describe a perturbation (changing flux) inducing a distributed response (eddy currents) that acts to stabilise the system against further change, dissipating energy in the process. The minus sign becomes a negative feedback loop. The derivative becomes a temporal sequence. The field relation becomes a causal chain.

Neither account is wrong. They are oriented towards different explanatory goals. The physicist seeks the minimal invariant encoding. The biologist seeks the maximal mechanistic unpacking.

The thesis of this paper is simple:

Physics retains what must be true. Biology reconstructs how it came to be true.

The consequence is that physics is amenable to mathematisation not because physicists are cleverer, but because the systems they study are closed enough to admit algebraic structure. When variables can be isolated, interactions are few-to-few, and conservation laws constrain the space of possibilities, description compresses to equation. When variables proliferate, interactions are many-to-many, and context dominates, description must expand into narrative. The six examples that follow trace this boundary.

Lenz's Law and Eddy Currents

The physicist's compression

Faraday's law of induction states:

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$

The minus sign is not a convention; it is the sign constraint required by energy conservation. Without it, induced currents would reinforce the change in flux, producing energy from nothing. The Poynting theorem provides the local energy balance:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$$

Eddy currents are simply the distributed realisation of this relation in conductors. The entire mechanism is implicit in the structure of the equations: direction, causation, constraint, dissipation.

The biologist's expansion

A biologist would not elevate the equation as the primary object. Instead, the account proceeds as a process:

A changing magnetic environment induces circulating currents that act to stabilise the system against flux change, dissipating energy as heat.

The minus sign becomes *negative feedback*, directly analogous to homeostasis or feedback inhibition in metabolic pathways. The flux derivative becomes a perturbation triggering a distributed response. The emphasis shifts from invariance to stability under perturbation, and from algebraic necessity to thermodynamic cost. The biologist asks: what is the conductivity? What geometry allows loops to form? What breaks first under perturbation?

The contrast

Aspect	Physics	Biology-style
Core object	Equation	Process
Minus sign	Algebraic necessity	Negative feedback
Eddy currents	Solution of Maxwell's equations	Distributed response mechanism
Emphasis	Invariance	Stability under perturbation
Energy	Often secondary	Central (dissipation, cost)

Mass Defect and Binding Energy

The physicist's compression

The entire nuclear binding story collapses into:

$$\Delta m = Zm_p + Nm_n - m_{\text{nucleus}}, \qquad E_b = \Delta m\, c^2$$

The "missing" mass is not missing; it is energy already released during formation. The strong interaction lowers the system's energy; the bound system sits in a potential well. Conservation of energy and relativistic mass–energy equivalence close the system. The equation *is* the mechanism, once written at the right level of abstraction.

The biologist's expansion

A biologist reconstructs a process: separate nucleons interact via short-range attractive forces, rearrange into a more stable configuration, and release energy to the surroundings, leaving a lower-energy bound structure. The scalar E_b becomes a transition between states. The equality $E_b = \Delta mc^2$ becomes a question about pathways: gamma emission,

kinetic recoil, intermediate configurations. The ground state becomes a position in a stability landscape, analogous to a protein folding funnel or a fitness landscape. The emphasis shifts from the invariant quantity to the architecture of the system: proton-to-neutron ratio, shell effects, magic numbers.

The contrast

Mass defect is particularly revealing because it *looks* like a static discrepancy (a number), but actually encodes a historical process (energy release during formation). Physics keeps only the invariant residue. Biology would reconstruct the history and pathway that produced it.

Specific Heat Capacity of Water

The physicist's compression

Water's anomalously high specific heat capacity ($c \approx 4.18\ \mathrm{J\,g^{-1}K^{-1}}$, roughly twice that of ice and four times that of most liquids) is encoded in the microscopic partition of energy. Heat capacity measures the number of accessible degrees of freedom weighted by their coupling strengths:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

For water, the hydrogen bonding network provides an extensive reservoir of intermolecular modes. Energy added as heat is absorbed not merely by translational and rotational kinetic energy of individual molecules, but by the excitation, bending, and partial disruption of an extended, fluctuating hydrogen-bond network. The high c is a consequence of the large number of coupled degrees of freedom per unit mass.

The biologist's expansion

A biologist would describe water's thermal inertia as a *functional property* of profound ecological consequence. Oceans buffer planetary temperature. Organisms exploit water's thermal stability to maintain homeostasis. Blood, lymph, and cytoplasm serve as thermal reservoirs that resist rapid temperature fluctuation.

The hydrogen bond network becomes an adaptive infrastructure: a cooperative, dynamic lattice that absorbs perturbation. The biologist asks not "how many degrees of freedom?"

but "what does this stability enable?" The answer is: thermal homeostasis at every scale from the cellular to the planetary.

The contrast

The physicist compresses to a derivative of internal energy with respect to temperature, explained by the density of accessible modes in the hydrogen bond network. The biologist expands to a functional narrative about what the property affords: buffering, stability, the thermal scaffolding of life. Same anomaly; different explanatory orientation.

Double-Slit Interference

The physicist's compression

The fringe pattern from a double slit is encoded completely in the superposition principle and the geometry of path difference. For two slits separated by d, the condition for constructive interference at angle θ is:

$$d \sin\theta = n\lambda, \qquad n = 0, \pm 1, \pm 2, \ldots$$

The intensity distribution for slits of finite width a combines single-slit diffraction with double-slit interference:

$$I(\theta) = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2 \cos^2\beta, \qquad \alpha = \frac{\pi a \sin\theta}{\lambda}, \quad \beta = \frac{\pi d \sin\theta}{\lambda}$$

The entire fringe pattern, including the positions, spacings, and relative intensities of all maxima and minima, is determined. There is no narrative. There is geometry, superposition, and λ.

The biologist's expansion

A biologist encountering this phenomenon would describe a process: a wave (or particle) encounters a barrier with two openings; it passes through both simultaneously (or, in the quantum case, the probability amplitude does); the two contributions recombine, and where they arrive in phase, the signal is strong; where out of phase, the signal cancels. The fringe pattern is the *record* of this recombination.

The biologist would emphasise the conditions: coherence of the source, relative scale of slit separation and wavelength, the detection surface. They would note that the pattern is fragile, context-dependent: change the slit width, the wavelength, the coherence length, and the pattern degrades or vanishes. It is not a universal law but a *regime-specific outcome* of an interaction between source, barrier, and detector.

For a quantum biologist, the single-particle version is even more striking: the pattern builds up statistically, one detection at a time, and disappears if "which-path" information is available. The biologist sees not an equation but a *conditional process*: outcome depends on what is measured, what is accessible, what interacts with what.

The contrast

The physicist: $d \sin\theta = n\lambda$. A complete, closed, predictive encoding. The biologist: a conditional process sensitive to coherence, geometry, and observation. Same fringes; different explanatory furniture.

Free-Fall Equivalence

The physicist's compression

That all objects fall with the same acceleration in a gravitational field, regardless of mass, is one of the most informationally dense compressions in physics. Newton's second law and the law of gravitation give:

$$ma = \frac{GMm}{r^2} \qquad \Longrightarrow \qquad a = \frac{GM}{r^2}$$

The mass m cancels. This is not a coincidence; it reflects the equivalence of inertial and gravitational mass, the observation that Einstein elevated to a principle and that became the foundation of general relativity. In the geometric formulation, free fall is not "acceleration under a force" but geodesic motion in curved spacetime. The universality is built into the geometry: all objects follow the same geodesics because gravity is not a force acting on mass but the curvature of the arena in which mass moves.

The biologist's expansion

A biologist, presented with the claim that a feather and a hammer fall identically (in vacuum), would immediately ask about the conditions. What medium? What drag? What Reynolds number? The universality is real but is the universality of an idealised limit. In any actual biological context, air resistance, buoyancy, terminal velocity, and morphology dominate.

The biologist sees falling not as a test of equivalence principles but as an *ecological event*: a seed disperses, a predator dives, a fruit detaches. The "acceleration" is the beginning of a trajectory shaped by drag coefficient, cross-sectional area, surface texture, and medium density. A maple samara autorotates; a peregrine falcon tucks its wings; a spider pays out silk to catch the wind. Each is "falling," but the physics of equivalence is the least interesting part of the story.

The biologist's instinct, that the conditions under which the clean cancellation holds are themselves the interesting question, is not wrong. It is simply oriented toward a different explanatory target: not the invariant, but the departures from it that structure real-world outcomes.

The contrast

The physicist compresses to a cancellation (m drops out) and elevates it to geometric principle. The biologist notes the cancellation, then immediately asks what happens when you restore the complexity that was stripped away. The invariant is the starting point for one discipline and the idealised limit for the other.

Parallel and Series Circuits

The physicist's compression

Consider N identical bulbs of resistance R connected to a battery of emf ε (internal resistance neglected for clarity).

In **series**, the total resistance is $R_{\text{total}} = NR$. The current through the circuit is:

$$I_{\text{series}} = \frac{\varepsilon}{NR}$$

The total power delivered is:

$$P_{\text{series}} = \frac{\varepsilon^2}{NR}$$

Each bulb receives power ε^2/N^2R. Adding bulbs in series *reduces* total power and dims each bulb.

In **parallel**, the total resistance is $R_{\text{total}} = R/N$. The total current drawn is:

$$I_{\text{parallel}} = \frac{N\varepsilon}{R}$$

The total power delivered is:

$$P_{\text{parallel}} = \frac{N\varepsilon^2}{R}$$

Each bulb receives full voltage and dissipates ε^2/R. Adding bulbs in parallel *increases* total power linearly, draining the battery N times faster.

The entire story, including why parallel bulbs are brighter, why the battery drains faster, and by exactly how much, is contained in Ohm's law and the definition of power: $P = IV = I^2R = V^2/R$.

The biologist's expansion

A biologist would frame this as a resource distribution problem. The battery is a finite energy store; the bulbs are consumers; the wiring topology determines how the resource is partitioned.

In series: a single supply chain, each consumer drawing from the same diminished flow. Analogous to a linear metabolic pathway where flux through the chain limits throughput. Adding consumers in series increases total resistance to flow, reducing per-unit delivery.

In parallel: independent supply lines from a shared reservoir. Each consumer draws at full capacity. Analogous to capillary beds branching from an artery, or parallel metabolic pathways each accessing the same substrate pool. Total demand scales with the number of branches, draining the reservoir proportionally faster.

The biologist emphasises the trade-off: parallel access maximises per-unit output but accelerates resource depletion. This is a ubiquitous biological constraint, from nutrient allocation in root networks to oxygen delivery in circulatory systems.

The contrast

The physicist writes $P = N\varepsilon^2/R$ and the story is told. The biologist unfolds this into a resource allocation narrative: topology determines distribution, and distribution determines depletion rate. The equation is a compressed version of what the biologist would call a supply-and-demand architecture.

The Underlying Distinction

Across all six examples, the same pattern recurs:

	Physics	**Biology-style**
Primitive	Relations (symmetries, conservation laws)	Systems (components, interactions)
Direction	Compress to invariants	Expand to mechanisms
Mechanism	Implicit in mathematical structure	Explicit, decomposed, narrated
Mathematisation	Natural (low-dimensional, closed)	Difficult (high-dimensional, open)
Goal	Minimal sufficient description	Traceable causation across components

The reason physics is amenable to mathematisation is not merely that physicists prefer equations. It is that the systems they study admit closure: a finite number of variables, constrained by symmetries and conservation laws, interacting in ways that can be fully captured by algebraic or geometric relations. Under these conditions, a good description *is* already a mechanistic explanation. The equation carries the mechanism inside its structure.

Biological systems lack this closure. They are high-dimensional, open, path-dependent, and historically contingent. A gene regulation network, a metabolic pathway, an ecosystem: none of these can be compressed into a single invariant relation without losing precisely the information that matters. So biology explains by expanding: decomposing into components, tracing causal chains, specifying conditions, narrating processes.

The distinction is not description versus mechanism. It is that, in physics, at the fundamental level, the distinction collapses. In biology, it persists.

Conclusion

The axis of contrast is *closure*. In electromagnetism, nuclear physics, classical mechanics, and wave optics, the system is closed enough to admit algebraic structure. Description compresses to invariant. A cross product encodes a three-way interaction. A mass cancellation encodes geometric universality. A minus sign encodes conservation.

In biological and ecological systems, closure fails. Variables proliferate. Context dominates. History matters. So the same phenomena, when re-described in biological terms, necessarily expand into process, narrative, and conditionality.

Physics compresses to invariants. Biology expands to dependencies.

Neither mode is superior. They are adapted to different regimes of the same explanatory landscape.